

Lucia (Lucy) O'Toole

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EDUCATION

Worcester Polytechnic Institute *B.S. Mathematics, B.S. Computer Science*

Aug 2022 - Aug 2025

Dean's List (Fall 2022 - Fall 2024)

GPA: 3.79

Coursework: Graph Theory, Group Theory (Rubik's Cube), Real Analysis I/II, Graduate Linear Algebra, Operating Systems, Computer Networks, Machine Organization, Theory of Computation, Linear Programming

OBJECTIVE: Seeking a research engineering position where I can collaborate with other highly motivated and ambitious engineers and innovators.

PROFESSIONAL EXPERIENCE

Qualcomm Inc. - Software Engineer

Nov 2021-Jan 2023

- Applied network design optimization research to neural networks. Co-invented patent-pending system for parallelizing neural networks. Collaborated on network optimization techniques for neural network training.

Reservoir Labs Inc. - Engineering Staff

Jun-Nov 2021

- Conducted R&D for computer network design optimization. Created programs analyzing impact of fiber optic cable additions between datacenters. Company acquired by Qualcomm.

Pakira Inc. - Core Technical Team

Feb 2019-Jun 2021

- Technical Lead for Dataset Collection. Full-stack developer (web GUI, front/back-end, database, scripting).

RESEARCH EXPERIENCE

WPI Major Qualifying Project: "Perfect and Pretty Good State Transfer in Weighted Discrete Quantum Walks" Jan-May 2025

- Researching linear algebra, group theory, and graph theory methods for quantum walks on graphs. Applications in Quantum Computing research and development.

Published: <https://digital.wpi.edu/pdfviewer/vq27zs37q>

WPI Interactive Qualifying Project: "Web Scraping Needs and Limitations"

Jun-Aug 2024

- Researched web scraping methods and applications for automated data collection. Identified limitations through background research, interviews, and surveys. Developed recommendations for new technological features.

Published: <https://digital.wpi.edu/pdfviewer/6t053m080>

PATENTS

Patent Website: <https://patents.justia.com/inventor/lucia-regina-o-toole>

N. Sebastián, et al. "Systems and Methods for Parallelizing Neural Networks Using Bottleneck Structures"

Appl. 18/503,005 (Pending)

N. Amsel, et al. "Multi-Domain Network Data Flow Modeling"

Appl. 18/049,159 (Pending)

RESEARCH AREAS: Quantum Computing, Performance Optimization, Theory, Systems, Networks, Algorithms

TECHNICAL SKILLS: Java, Python, R, SQL, C++, OCaml, LaTeX

ACTIVITIES: Philharmonic Orchestra (Cello), Artisanal Bread Baker, SCUBA, Sailing

AWARDS: Merit. Math Award, Pi Mu Epsilon (Alpha), WPI Dean's List 22-24, Putnam Score 14, PROMYS Star of the Week